QP CODE:25900146

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATIONS MARCH 2025

Second Semester

MG2DSCMOS100-FOUNDATIONS OF MOBILE DEVELOPMENT SYSTEMS

2024 Admission Onwards

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

SCHEME OF EVALUATION

PART A

Multiple Choice Questions

Answer all questions. Each question carries 1 Mark

	Learning Domain	CO No(s)
1. b). list.remove(value);	U	1
2.c) break	U	1
3. a). Hiding the details and showing only the necessary parts.	U	1
4. a). Installing Flutter SDK	K	2
5.a). They automatically adjust their appearance based on the platform.	U	2
6. d). Configuration files	K	2
7. c). To retrieve data from a server	K	3
8. b). Cloud Firestore	U	3
9. c). SharedPreferences	K	3
10. b). Synchronize data changes across devices	U	3

PART B

Short Answer Questions

Answer 4 questions. Each question Carries 5 Marks

11. Explain the difference between Set and List in Dart. U 1

Definition of Set and List-2 Mark

Comparison- 3 mark

12. What is exception handling, and why is it important in programming? Explain with an example **K 1**

Definition of Exception Handling -1 Mark

Importance of Exception Handling -2 Mark

Example and Explanation of Example-2 mark

13. How does Shared Preferences store data in a Flutter application? U 2

Explanation-5 Mark

14.**What is the key difference between a Stateless and a Stateful widget in Flutter?** **K 2**

Definition of Stateless Widget and Stateful Widget -2mark

Key Difference-2 mark

Example -1 mark

15. What are the differences between Firebase Authentication and Firebase Firestore in Flutter? **U 3**

Differences -3 mark

Example of each-2 mark

16. What is the purpose of the DELETE method in RESTful APIs? K 3

Explanation-5 mark

PART C

Essay Questions

Answer any 2 Questions. Each carries 10 Marks

17. Discuss the advantages and disadvantages of using Lists, Sets, and Maps in Dart. **K 1**

Advantages of Lists, Sets and Maps-5 Mark

Disadvantages of Lists, Sets and Maps -5Mark

18. Explain the concept of classes in Dart. How do they help in organizing and structuring Flutter applications? **K 1**

Concept of classes-4mark

Explanation-3 Mark

Syntax-3 Mark

19. Explain the role of setState() in Flutter state management. How does it impact UI updates? Provide examples. **U 2**

Definition -4 Mark

Explanation-3 Mark

Example-3mark

20. How does Flutter's reactive UI system differ from traditional imperative UI frameworks? U 3

Definition of Reactive UI and Imperative UI framework -4 mark

Comparison-6 Mark